

Research Assignment 1
Course: Data Analytics
Beginner Students
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1. Database: also referred to as a relational database, database is an organised collection of information capture in real-time. usually stored in tables with rows and columns. Data bases are essential in the modern world because they are used to power up applications or manage data in organisations. ~~The following are two examples of database~~
e.g. Computer, facebook
databases are usually used in retail stores and Hospitals.
2. Data warehouse: is a database or used to store multiple data bases. Data warehouse is usually used for reporting and analytical process for decision-making purposes. Unlike databases, the data warehouse consists of already summarised data extracted the ETL process therefore the data is specific and is for specific purposes. Data warehouse is not real time data, does not interfere with the process and it is usually faster.
A company would use a data warehouse because it is capable of storing extremely large data.
3. Data lake: Captures all types of data in which can be stored in a data, usually captures large raw data. Data lake differs from data warehouse because it requires that data to be still cleaned, while data warehouse consists of summarised data. Data lake data cannot be used for reporting and analysis purposes.

Data lake is useful for companies than the datawarehouse because of the costs of storing data.

4. Data Mart: it consists of subsets of datawarehouse. The data can be summarised for specific purpose for particular departments. The data is retrieved and relieved for specific teams (Simplified data for crucial information)
It extracts a particular amount of data from the data warehouse
5. -Structured data is data that is organised and comes in a consistent format. usually stored in tables with columns & rows. ~~and~~ example: Spreadsheets
-Unstructured data is not consistent and therefore does not have a specific single format. Such as: PDF documents or emails
-Semi-Structured. it is a combination of both structured and unstructured data For example web, Emails & XML
6. Data Modeling Data Modelling is the process of designing how data is structured and Related, it ensures efficiency and retrieval. The two types of data modelling is: Conceptual model (high level) and logical model (detailed structure)
7. Data Mining: it is a process of finding patterns in large datasets. Usually business use it to recommend products or detect fraud. e.g In Retail such patterns can determine consumer patterns and therefore can recommend consumer preference. In banking they can help detect and deter fraud.

~~8. DBMS~~

8. DBMS (Database Management Systems) are softwares used to manage databases.

four types of DBMS platforms

- * MySQL = it is usually used to power up web applications
- * PostgreSQL = it is used for complex analytics.
- * Oracle = it is used for enterprise systems
- * MongoDB = it is more flexible although does not consist of SQL data.

9. ETL (Extract, transform, load) - it's a process that transfers data ^{and from} to the data warehouse and ensures that data is usable and accurate. Extract - collects the data, Transform: Cleans and processes the data and Load: stores the data in the system.

10. Data storage formats:

- * CSV - it is simple tables used to store data (comma separated values)
- * JSON - These are APIs and data exchange. (JavaScript Object Notation)
- * Parquet - it is an open-source file format designed for efficient data storage and retrieval
- * Avro - it is a binary oriented format usually used for data streaming

Avro format is usually used in organisations for operational system, the parquet format is usually for analytical purposes and data is efficiently summarized. JSON is used to store and transmit data between a web server and client and stores user submission forms. CSV is to both move and update data into other programs efficiently.

SECTION B: Artificial Intelligence and Machine Learning.

11. AI: AI is the ability of machines to perform tasks that require human intelligence. there are two types of Artificial intelligence which the Narrow AI and General AI.
Narrow AI (weak AI) focuses on a narrow, single subject tasks for example: Self driving cars, Siri and Social Media notifications.
General AI (AGI) or Strong AI are developed to perform human like tasks such as predictive weather or physical trainer (not yet fully developed).
12. Machine learning (ML) is a subfield of Artificial Intelligence (AI) it allows systems to learn from data instead of being explicitly programmed. There are three types of ML Supervised, unsupervised and Reinforcement.
Supervised is labeled data, unsupervised ^{finds} ~~uses~~ patterns (un-labeled) whiles reinforced learns through rewards.
13. Deep learning: it is a subfield of machine learning of which is a subfield of AI. It learning algorithms that accept data as inputs, processes and deliver an output.
Neural Networks are systems modeled after the human brain that processes data in layers.
14. What is NLP: it is Natural language programming, it enables machines to understand human language. the three real-world examples consist of Chatbots, translation apps and voice assistants.

15. LLM: It is a large language model which processes and generates human like text. Machine learning work best with structured data while LLMs are trained on massive unstructured text datasets. The two main known LLMs are ChatGPT and Claude, chat gpt processes through conversations while Claude is mainly used for writing and analysis.
16. Generative AI creates new content such as text, images etc. It summarises complex information and generate new ideas. The generative tools consist of ChatGPT - text, DALL-E - images and Midjourney for art.
17. Training datasets are used to train AI models how to perform tasks. The quality of training data is important because inaccurate or biased data can lead to incorrect or unfair results.
18. Computer vision is a field of AI that enables machines to interpret and understand visual information from images or videos. Machines "see" by analysing pixel data and recognising patterns. Two examples are: facial recognition systems and Self-driving cars detect objects.

SECTION C: APIs, Tools & Developer concepts

19. API: Application Programming interface is set of rules that allows different software systems to communicate and share data. Analogy: It is like a waiter taking your order and delivering it from the kitchen.

20. An API key is unique identifier used to authenticate and authorise access to an API. It must be kept secure because if exposed, unauthorised users can misuse the service.

21. REST stands for Representational State Transfer.

Methods:

- GET: retrieve data
- POST: send new data
- PUT: update existing data
- DELETE: remove data

22. JSON is a lightweight format used to store and exchange data. example:

```
{ "name": "Millicent",  
  "age": 25,  
  "city": "Maputo"  
}
```

It is widely used because it is easy for humans and machines to read.

23. Prompt Engineering is the process of designing effective inputs to get better responses from AI systems. a poorly written prompt would be vague such as "Explain AI" better would be a more descriptive prompt. "Explain AI in simple terms with real-world examples for beginners".

24. MAP Function applies a specific operation to each item in a list and returns a new list. example (python)

```
numbers = [1, 2, 3]
```

```
result = list (map (lambda x: x * 2, numbers))
```


25. Cloud Computing: Cloud Computing refers to using remote servers on the internet to store and process data instead of local machines

- IaaS: Provides infrastructure
- PaaS: Provides development platform
- SaaS: Provides ready-to-use software

26. Version Control: is a system that tracks changes to files or code over time. Git is widely used because it allows collaboration and version tracking

Commit: Saving changes

Branch: Creating a separate version

27. Jupyter Notebook: It is an interactive environment used for writing and executing code, especially for data analysis. It is commonly used with python and allows combining code, text and visuals.

28. Python is a high-level programming language widely used in data analytics and AI because it is simple and powerful.

Libraries:- Pandas = used for data manipulation

- NumPy = numerical operations
- Matplotlib = visualization
- Scikit-learn = machine learning.

Section D: Ethics & Careers

29. Data privacy refers to protecting personal information from unauthorised access. GDPR is a regulation in the European Union that protects user data and applies to organisations handling EU citizens data.

30. Data Ethics involves using data responsibly and fairly. three examples of unethical uses of data are:

- Using personal data without consent
- Biased AI decisions
- Lack of transparency in algorithms

31. Data governance is the management of data availability, usability, security and integrity. The key components of a good data governance framework are: Data quality, security policies, and compliance rules.

32. Bias in AI occurs when a model produces unfair results due to biased data. This can lead to discrimination and inaccurate outcomes.

33. Role	Focus	skills	Tools
Data Analyst	Analyze data	Excel, SQL	Power BI
Data Scientist	Build models	Python, ML	TensorFlow
Data Engineer	Build pipelines	SQL, coding	Hadoop
AI Engineer	AI Systems	Deep learning	PyTorch

34. What is BI? Business Intelligence involves analysing data to support business decision. Tools like PowerBI and Tableau help visualize data and identify trends.

SECTION E: Reflection (Expanded)

35. The concept that was most surprising in this assignment is API and API key and the one I found most interesting is the difference between AI, deep learning and Machine Learning

36. Deep learning was difficult because it involved complex concepts like neural networks. I used online videos such as youtube to better understand the concept.

37. Data and AI will play a major role in my career by helping me analyse data, make informed decision and improve efficiency. For example, I can use data tools in business or research roles.

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